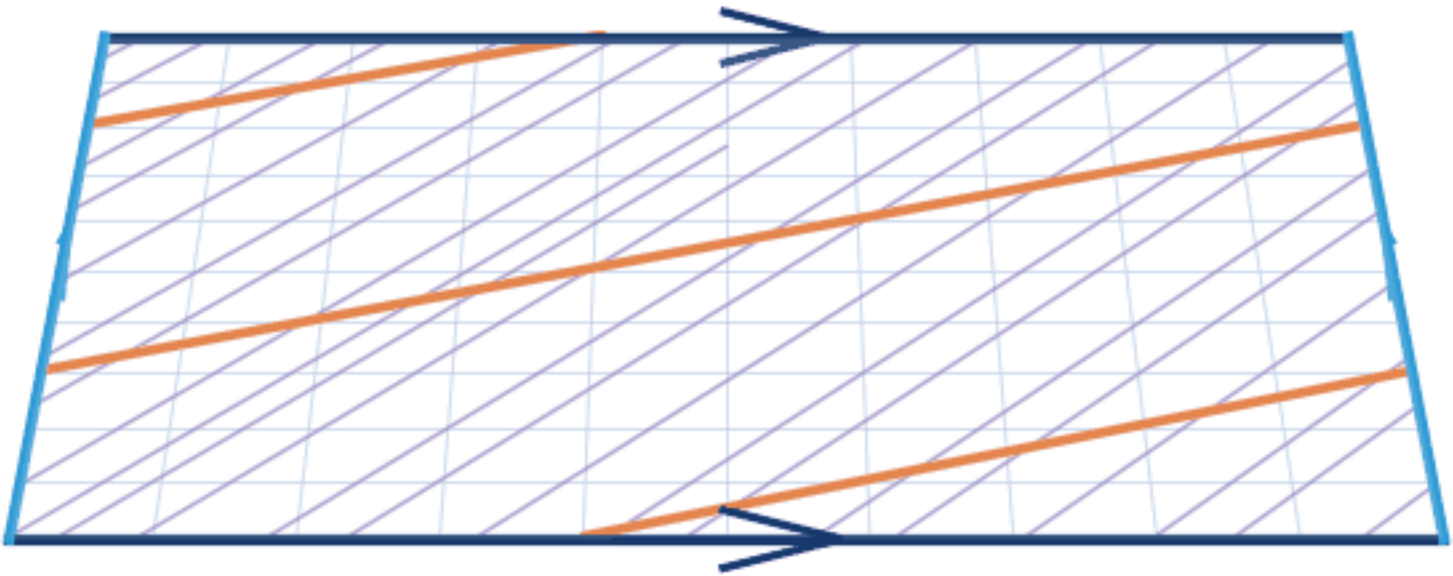
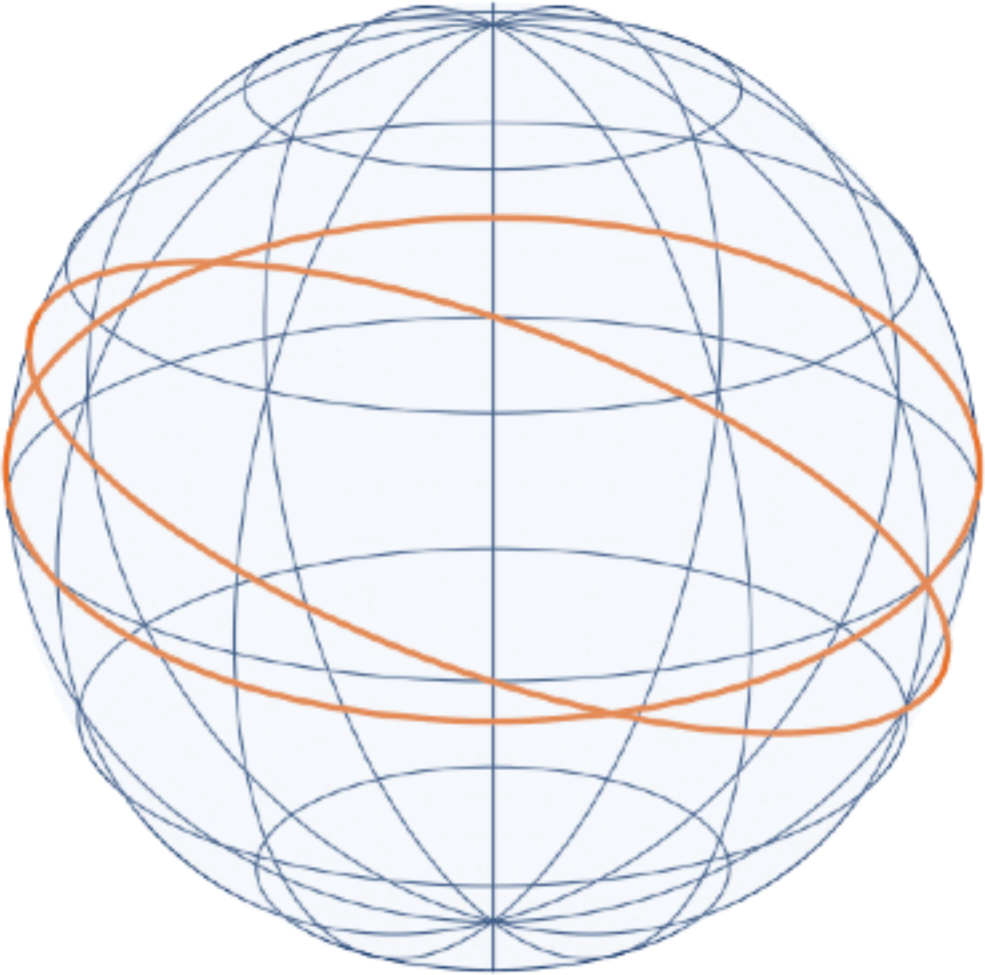
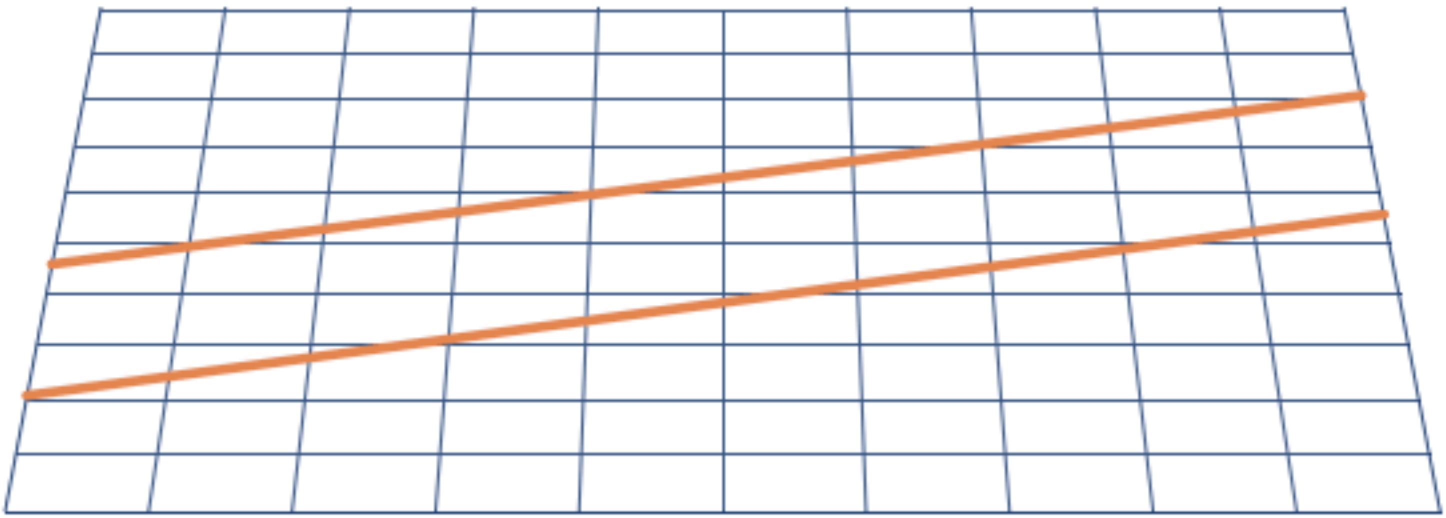






ISOTROPIC METRICS

On familiar spaces, distance looks the same in every direction — and the same at every point.

Round sphere

geodesics: straight lines

Flat torus

Euclidean plane

$$\mathbb{R}^2$$

$$\mathbb{S}^2$$

$$\mathbb{R}^2/\mathbb{Z}^2$$

geodesics: great circles

$$ds^2 = dx^2 + dy^2$$

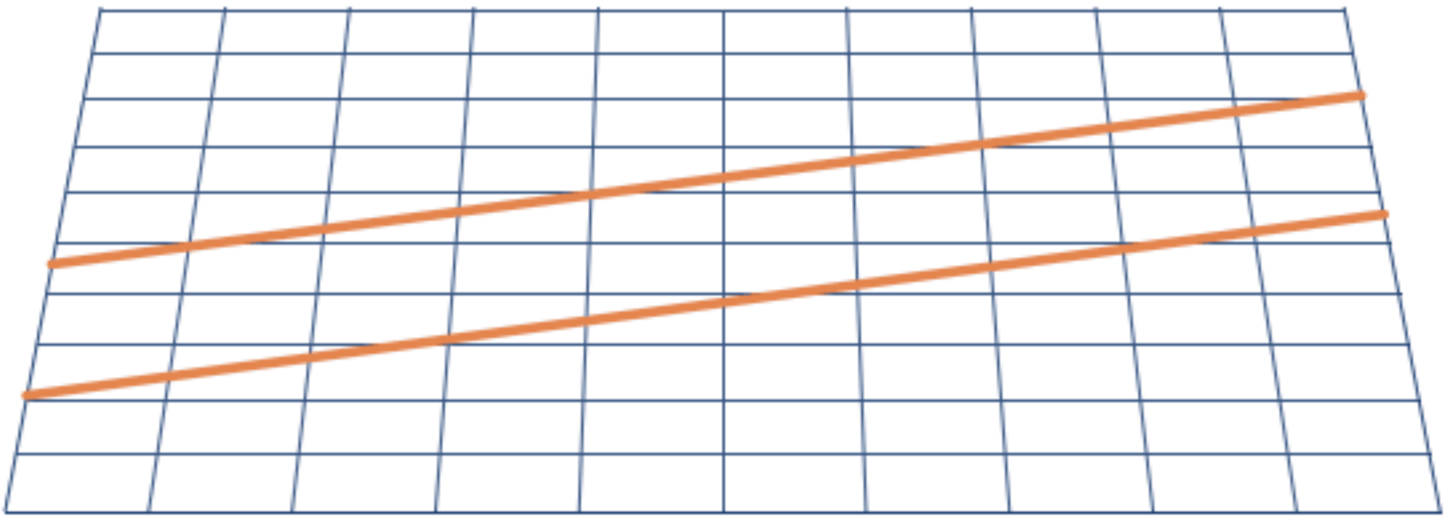
$$ds^2 = d\theta^2 + \sin^2\theta \, d\varphi^2$$

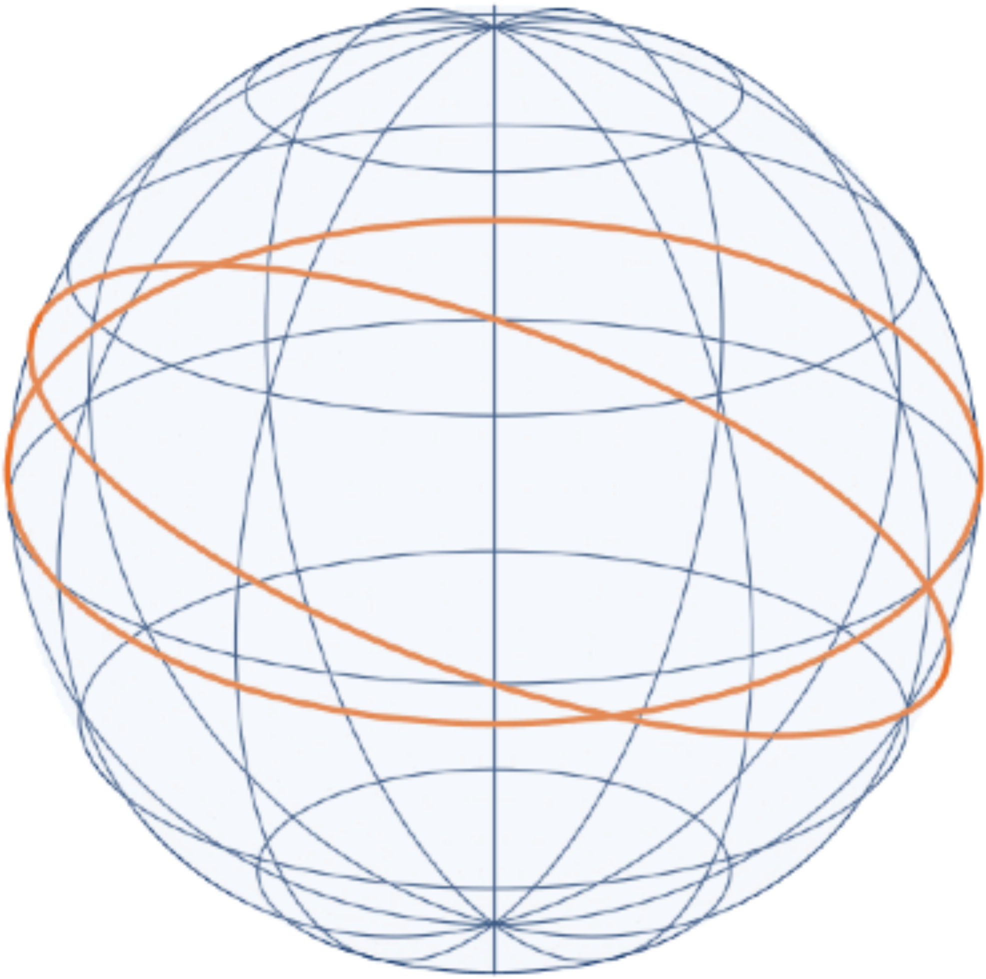
$$ds^2 = dx^2 + dy^2$$

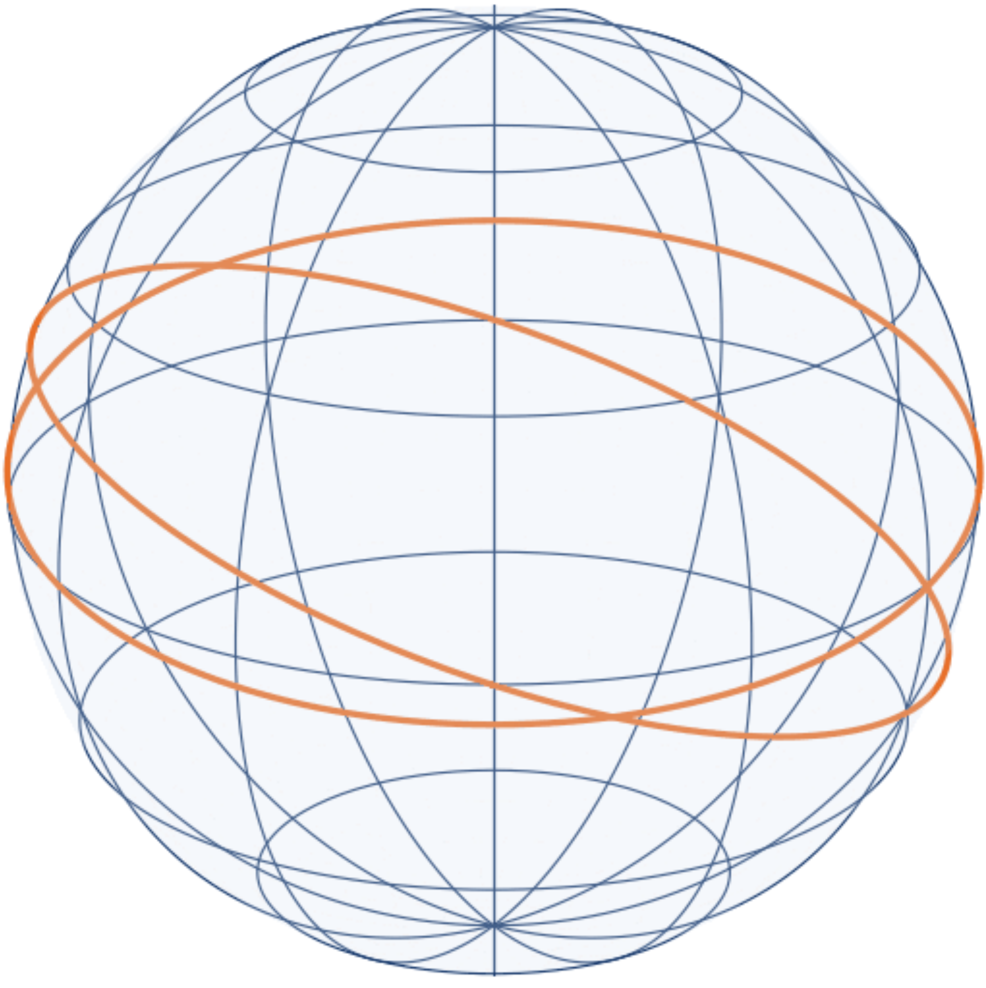
geodesics: straight lines

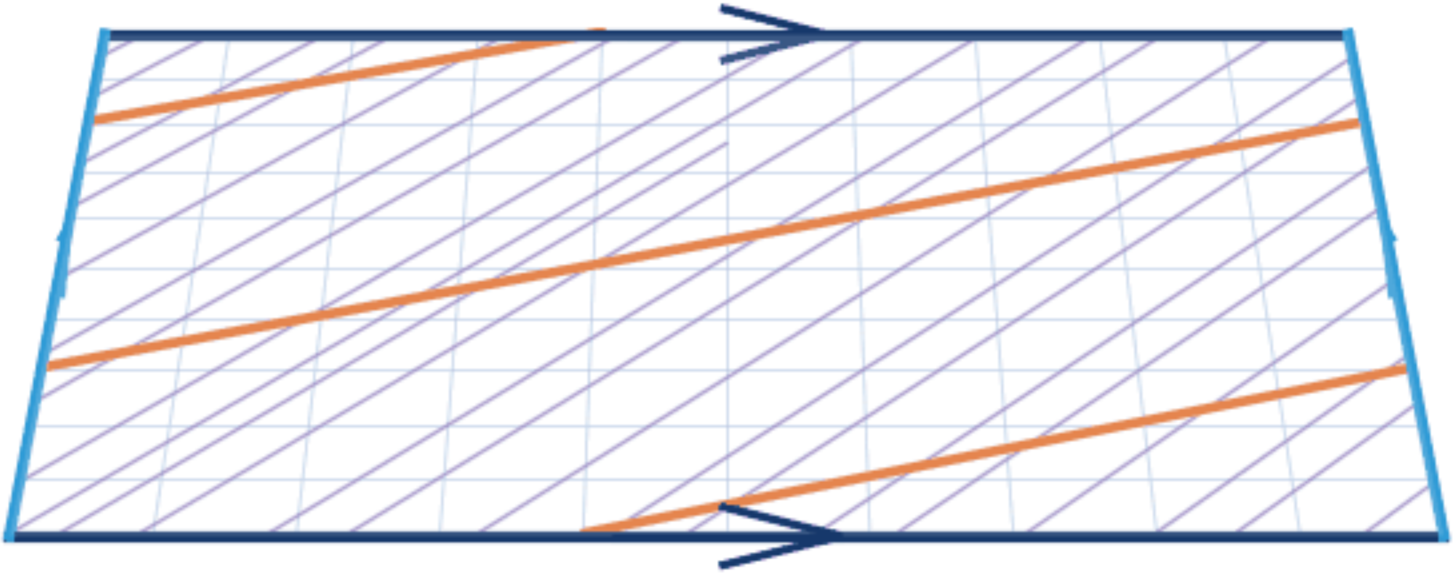
(locally Euclidean)

05



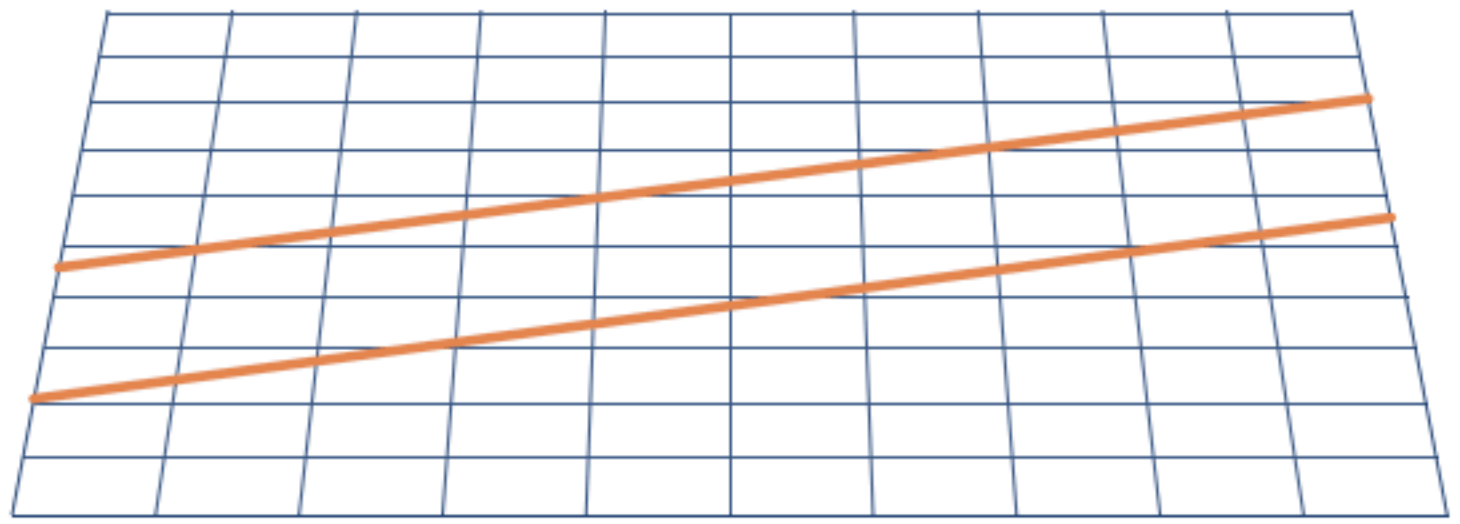






On familiar spaces, distance looks the same in every direction — and the same at every point.

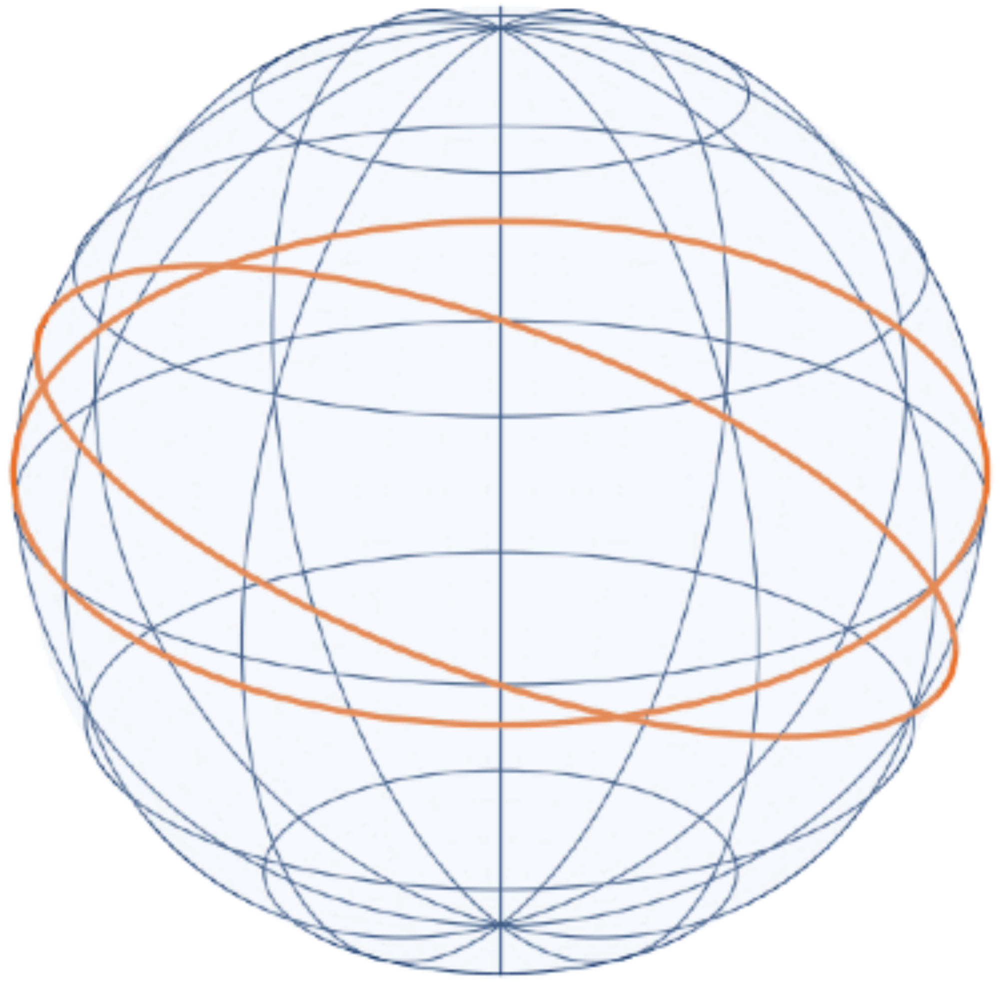
Euclidean plane $\mathbb{R}^2$



$ds^2 = dx^2 + dy^2$

geodesics: straight lines

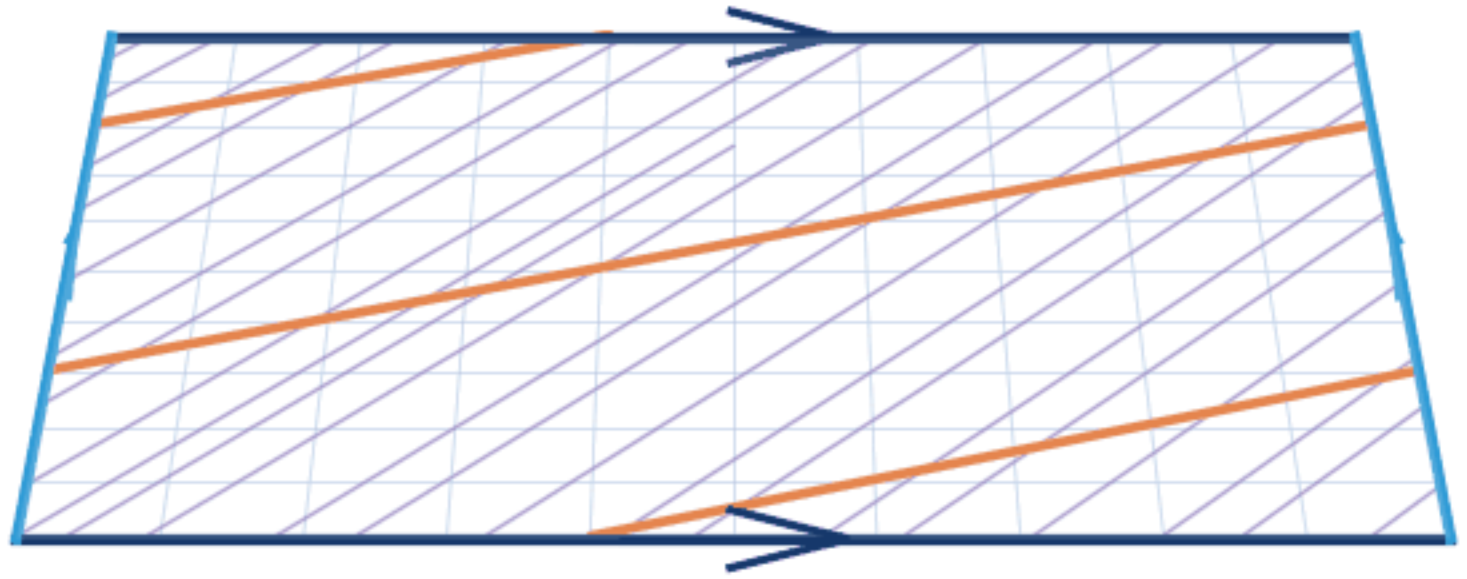
Round sphere $\mathbb{S}^2$



$ds^2 = d\theta^2 + \sin^2\theta\, d\varphi^2$

geodesics: great circles

Flat torus $\mathbb{R}^2/\mathbb{Z}^2$

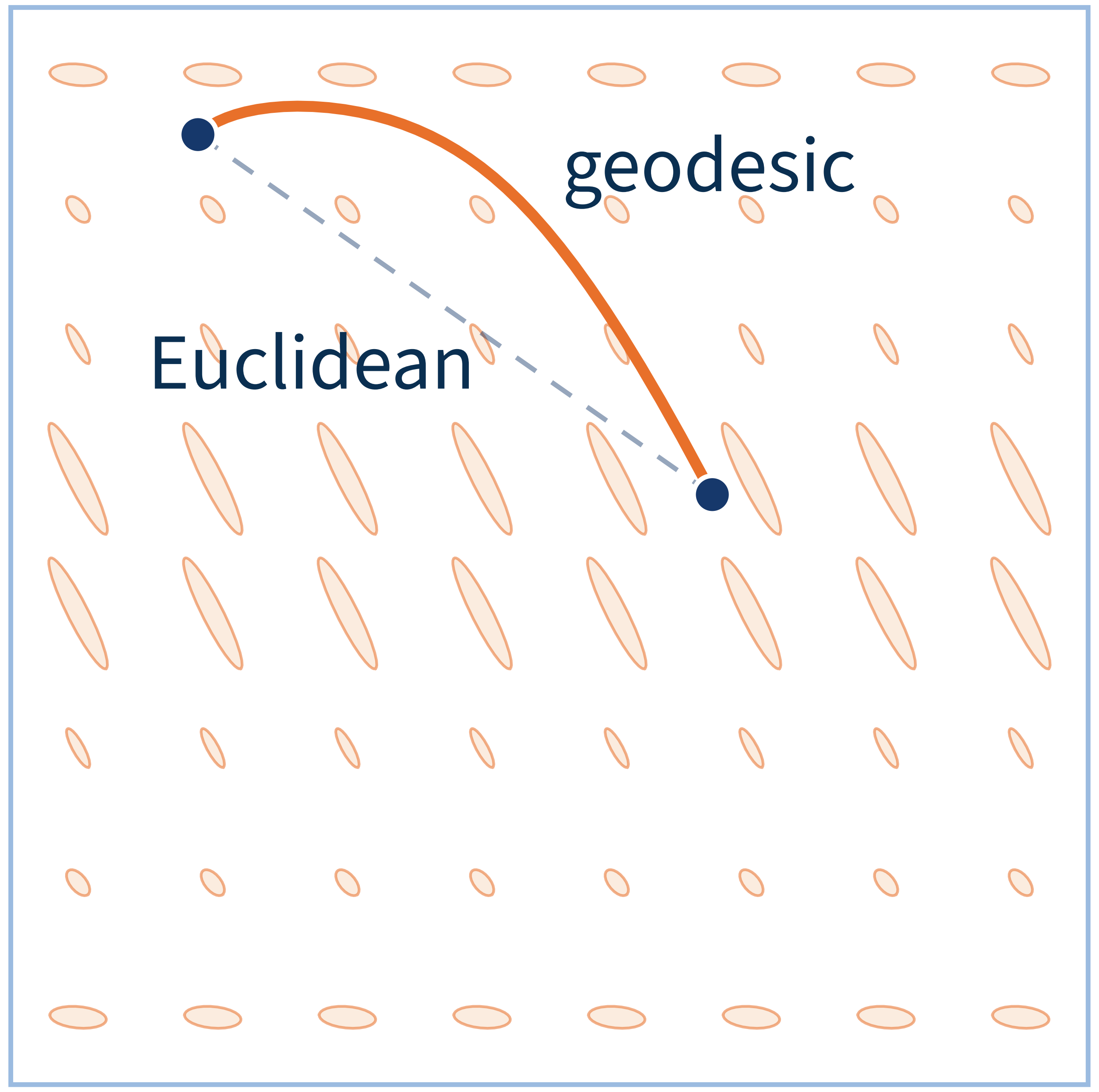


$ds^2 = dx^2 + dy^2$ (locally Euclidean)

geodesics: straight lines

Distance now depends on *where* you are in configuration space — some directions cost more to move through than others.

Configuration space



pullback

$$G(q) = J^{\mathsf{T}} \mathbf{I} J = J^{\mathsf{T}} J$$

Task space

> Pull the task-space metric back into configuration space
> — a straight line in task space becomes a *geodesic* in configuration space